NOT MEASUREMENT SENSITIVE
MIL-P-85856/40
7 December 1988

MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS TUNGSTEN CARBIDE-COBALT, TYPE II, CLASS 2, COMPOSITION J

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition

Fe	C	Co	W
1.5 max.	5.15	11.5 13.0	79.5 min.

Particle Size/Distribution

+200 Mesh (+75 µm)	1.0 max.
-270 Mesh (-33 µm)	95.0 min.
Mesh (-15 µm)	15.0 min.

Metallagrophic Inspection

When examined at 500x magnification, the tungsten carbide particles shall be capable of exhibiting a uniform distribution of crystallites, predominately in the range of 2 to 5 micron and not exceeding 20 microns.

Sprayed Coating Properties

Bond Strength	45.0 MPa (6500 psi) min.
Hardness	Rockwell 15N, 85 min.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 3% max. Unmelted Particles - 1% max.

AMSC N/A FSC 3439
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Deposit Efficiency and Deposition Rate

60% min for A/H. @ 7.0 lb/hr

Comparable Documents

Powder which meets the requirements specified is qualified under the following specifications.

General Electric Company	B50TF27, Class A
Aerospace Material Specification	AMS 7880

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/40)

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